

YUMA 27 ENE, AZ, USA

WMO: 722790

Lat: 32.8350N

Lon: 114.1884W

Elev: 620

StdP: 14.37

Time Zone: -7.00 (W07)

Period: 08-19

WBAN: 53154

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	36.0	39.4	-5.7	4.1	70.7	-0.8	5.4	67.1	17.0	57.8	14.9	58.6	2.0	N/A	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	110.4	68.4	108.4	68.3	106.3	68.0	75.3	92.3	74.4	92.9	73.5	93.7	7.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	117.3	82.4	69.1	109.5	83.3	67.0	101.6	85.9	39.0	91.8	38.2	91.9	37.3	92.9	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.6	13.5	11.8	DB	28.7	115.3	3.5	2.5	26.1	117.1	24.1	118.5	22.2	119.9	19.6	121.7	(4)
(5)			WB	23.0	77.5	2.8	1.3	21.0	78.4	19.3	79.1	17.7	79.8	15.6	80.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.0	56.6	59.2	66.1	71.5	77.3	88.1	93.4	93.0	87.2	75.4	64.2	54.8		
	DBStd	14.61	5.52	6.70	6.25	6.01	6.36	5.35	3.88	3.96	5.13	6.01	6.21	5.62		
	HDD50	42	15	7	0	0	0	0	0	0	0	0	1	19		
	HDD65	933	261	180	63	15	2	0	0	0	0	6	90	316		
	CDD50	8794	221	265	500	644	845	1142	1346	1333	1116	786	426	170		
	CDD65	4210	2	18	99	209	382	692	881	868	666	327	65	1		
	CDH74	68043	72	348	1535	3125	5816	11777	15231	14821	10268	4150	857	43		
	CDH80	43900	3	66	546	1473	3352	8122	10878	10490	6595	2110	264	1		
(14) Wind	WSAvg	5.0	4.3	4.4	5.2	6.3	6.2	5.8	5.9	5.4	4.7	4.1	3.9	4.1		
(15) Precipitation	PrecAvg	3.8	0.5	0.6	0.4	0.1	0.0	0.0	0.2	0.4	0.5	0.2	0.2	0.5		
	PrecMax	7.9	2.8	1.9	2.0	1.1	0.6	0.2	0.8	1.0	3.1	1.8	1.3	2.1		
	PrecMin	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.0	0.7	0.6	0.5	0.2	0.1	0.0	0.2	0.3	0.7	0.4	0.3	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.3	85.8	93.8	99.1	105.1	113.5	113.2	112.3	108.2	99.5	90.1	78.1		
		MCWB	52.0	54.5	58.2	59.1	61.1	66.2	69.0	69.2	67.0	64.3	57.7	54.2		
	2%	DB	76.1	82.1	89.1	95.2	101.9	109.6	110.0	110.0	105.4	96.3	86.4	74.8		
		MCWB	50.9	53.9	56.5	58.2	60.6	65.6	69.1	69.8	67.2	62.2	56.7	52.5		
	5%	DB	73.0	78.4	86.0	92.0	98.4	107.0	107.7	107.8	103.7	93.8	83.3	70.9		
		MCWB	50.1	52.9	55.3	57.5	60.0	65.1	68.9	69.7	67.2	61.1	55.6	51.0		
	10%	DB	69.9	75.0	82.8	88.4	95.1	104.4	105.7	105.5	101.3	91.0	79.2	67.3		
		MCWB	49.1	52.1	54.2	56.2	59.3	64.1	68.8	69.5	67.7	60.7	54.5	49.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.2	59.3	59.3	61.9	64.0	71.7	75.8	76.8	76.1	71.1	62.1	58.8		
		MCDB	66.5	75.1	85.5	89.7	96.1	98.0	92.0	94.6	88.7	81.6	74.1	68.7		
	2%	WB	56.4	56.6	57.9	59.7	62.2	70.3	74.6	75.5	74.7	68.3	59.6	56.8		
		MCDB	63.5	73.2	84.0	88.9	94.8	97.0	93.5	93.0	89.5	85.8	77.1	67.8		
	5%	WB	53.8	54.9	56.7	58.7	61.0	68.5	73.8	74.8	73.5	66.0	57.9	54.8		
		MCDB	64.3	72.0	82.4	87.2	92.9	98.2	94.7	93.5	89.6	84.9	77.2	66.3		
	10%	WB	51.7	53.4	55.3	57.5	59.9	66.4	72.9	73.9	72.3	64.0	56.2	52.7		
		MCDB	65.8	71.6	80.1	84.7	91.0	99.5	95.3	94.1	90.3	83.4	75.9	63.8		
(35) Mean Daily Temperature Range	MDBR		24.5	25.1	27.2	27.9	28.1	29.6	23.4	23.7	24.7	26.7	25.7	22.9		
	5% DB	MCDBR	29.8	30.4	31.9	33.3	33.2	31.3	26.0	26.1	27.7	30.0	30.3	27.5		
		MCWBR	14.1	13.2	12.6	13.0	12.4	11.2	8.0	8.1	9.2	12.2	13.7	13.7		
	5% WB	MCDBR	20.8	25.2	29.1	29.1	30.3	26.4	19.9	20.9	20.2	22.0	25.4	21.5		
		MCWBR	10.8	11.1	11.9	11.7	11.2	10.4	6.7	7.0	8.4	10.0	13.2	11.5		
(40) Clear-Sky Solar Irradiance	taub		0.283	0.287	0.306	0.319	0.330	0.337	0.381	0.378	0.341	0.315	0.291	0.283		
	taud		2.535	2.540	2.456	2.425	2.413	2.404	2.393	2.387	2.463	2.486	2.528	2.548		
	Ebn at Noon		300	310	309	307	303	299	285	285	293	295	294	294		
	Edh at Noon		26	29	34	36	37	38	38	37	33	30	26	24		
(44) All-Sky Solar Radiation	RadAvg		1125	1440	1898	2326	2585	2672	2400	2233	1978	1604	1223	1017		
	RadStd		54	92	62	55	59	92	83	83	64	60	50	52		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	-2.81	N/A	-0.63	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page